

Mid-term examination on Logic Circuit Design (Group 2)

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1 Balanced Ternary

Question. Write a C function which converts an integer of type `int` into its balanced ternary representation as an array of `int`. The prototype is

```
int* from10to3b (int dec);
```

The result of the function is an array of integers where the first element is the size of the balanced ternary representation and the remaining elements are the digits (also known as *trits*), the least significant being the last. For example, `from10to3b(17)` returns an array

4	1	-1	0	-1
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because $17_{10} = 1\bar{1}0\bar{1}_{3b}$.

Hint. First, compute the length of the balanced ternary representation by using the same algorithm used for the conversion. Then allocate an array and finally run the conversion algorithm as usual. Try some examples, like -17 . In C, the sign of `x % y` is the sign of `x`. You need to include `stdio.h` and `stdlib.h` (for `malloc`).

Answer.

```
// This program converts a number from its decimal representation
// into its balanced ternary representation.
```

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
```

```
int* from10to3b (int dec) {
```

```

int index = 0;
int input = dec;
int rem; // Safe because followed by a 'do' loop where it is
          // assigned a value.
do {
    rem = dec % 3;
    dec /= 3;
    if (rem == 2) {rem = -1; dec++;}
    else if (rem == -2) {rem = 1; dec--;}
    index++;
} while (dec != 0);
int* bter = (int*) malloc((index+1) * sizeof(int));
bter[0] = index;
do {
    rem = input % 3;
    input /= 3;
    if (rem == 2) {rem = -1; input++;}
    else if (rem == -2) {rem = 1; input--;}
    bter[index] = rem;
    index--;
} while (input != 0);
return bter;
}

void print (int* num) {
    for (int index = 1; index <= num[0]; index++)
        if (num[index] == -1)
            printf("I");
        else printf("%d",num[index]);
    printf("\n");
}

int main(int argc, char** argv) {
    if (argc == 2) {
        int num = 0;
        for (int i = 0; i < strlen(argv[1]); i++)
            if ('0' <= argv[1][i] && argv[1][i] <= '9')
                num = num * 10 + (int)(argv[1][i]) - 48;
            else {
                fprintf(stderr,"Only decimal digits allowed.\n");
                return -2;
            }
        int* bter = from10to3b(num);
        print(bter);
        free(bter);
        return 0;
    }
    fprintf(stderr,"Provide exactly one decimal number.\n");
    return -1;
}

```

}

Question. If an `int` is stored as a 2-complement 32-bit machine word, what are the smallest and largest numbers your program can convert?

Answer. The smallest number represented with 32 bits in 2-complement is -2^{31} (i.e., the leftmost bit is 1 and all the others 0) and the largest is $2^{31} - 1$ (i.e., the leftmost bit is 0 and all the others 1).